

BOM Site Handover Checklist

Quote: QT-COLLINS-E43DC5

Customer: Mr Collins

Project: Mr Collins — 40-Room Hotel (Off-Grid Solar)

Option: Off-Grid Hotel Package

System: off_grid — 110.55 kW, 194 panels

Date: 15 May 2026

Core equipment

☐	Description	Qty	Notes
	Longi 570W Panel	194.0	
	Energy Precision 10kW 30.0kW Inverter (×3 parallel)	3.0	
	Energy Precisions 16.0kWh Battery	21.0	
	Mounting structure (rails) (proforma rail sets)	122.0	
	PV mid / end clamp (each) (3.5× panels)	679.0	
	MC4 connector pair (~1.2 pairs per panel)	233.0	
	RR 10MM Copper Cable (100m coil) (AC feeder coils vs inverter kW)	6.0	
	Battery cable 50mm copper (per meter) (~0.9 m per kWh (large bank))	302.0	
	Battery cable 35mm copper (per meter) (supplementary 35 mm runs)	168.0	
	125A AC breaker (inverter 30.0 kW > 8 kW)	1.0	
	AC isolator switch 40A (AC isolator per inverter)	3.0	
	100A changeover (1 per inverter / genset path)	3.0	
	250A MCCB battery breaker (battery bank protection)	34.0	

Balance of system (catalog BOM)

☐	Item	Qty	SKU / usage
	AC isolator switch 40A	3.0	BOS-AC-ISO-40A — AC side disconnect at inverter for safe maintenance.
	DC isolator switch 32A	17.0	BOS-DC-ISO-32A — Lockable DC disconnect for maintenance — typically one per string or per inverter DC input.
	Earthing kit (rod, clamp, 16mm earth cable)	14.0	BOS-EARTH-KIT — Equipment bonding and array grounding per local wiring rules.
	DC/AC labeling & warning kit	1.0	BOS-LABEL-KIT — Circuit labels, arc-flash / DC warning stickers for commissioning sign-off.
	MC4 connector pair	233.0	BOS-MC4-PAIR — DC plug connections at modules and string ends; ~1 pair per module interconnection.
	PV mid / end clamp (each)	679.0	BOS-PV-CLAMP-EA — Secures modules to rails (mid + end clamps).

	PVC trunking 25mm (per metre)	291.0	BOS-TRUNK-25MM-M — Cable management for AC/DC runs on wall/facade.
	13-way breaker box	12.0	BOX-DB-13WAY — Main AC DB for breakers, SPD, and feeds.
	6-way breaker box	1.0	BOX-DB-6WAY — Small AC distribution enclosure.
	AC cable 6 mm ² (100m coil)	3.0	CAB-AC-6MM-100M — Branch AC wiring, DB to loads (smaller runs); supplements RR10 where needed.
	Battery cable 35mm copper (per meter)	168.0	CAB-BATT-35MM-M — Battery interconnect and inverter-battery DC runs (hybrid/off-grid).
	Battery cable 50mm copper (per meter)	302.0	CAB-BATT-50MM-M — Main battery bank feeders (≥15 kWh).
	Single core 6mm PV cable (100m coil)	9.0	CAB-PV-6MM-100M — Larger arrays / longer runs (>30 panels or >20 kW).
	RR 10MM Copper Cable (100m coil)	6.0	CAB-RR10MM-100M — AC feeder from inverter to DB / changeover (heavy residential & commercial).
	Miscellaneous (larger kit — Option 2 reference)	1.0	KIT-MISC-BOS-L — Same as KIT-MISC-BOS for commercial-scale jobs.
	Mounting structure (rails)	122.0	MNT-RAIL-SET-350 — Roof rail segments + typical fixings per set — matches Ghana proforma 'Mounting Structure (Rails)'.
	100A AC breaker	1.0	PRT-AC100-1P — AC overcurrent protection at DB / inverter output.
	125A AC breaker	1.0	PRT-AC125-1P — Higher-rated AC protection when total inverter >8 kW.
	100A changeover	3.0	PRT-CO-100A — Grid / generator / inverter selection (hybrid & backup).
	63A DC MCB 2P	17.0	PRT-DC63-2P — DC string / combiner protection (one per string group).
	250A MCCB battery breaker	34.0	PRT-MCCB-250A — Battery bank main overcurrent protection.
	SPD / lightning arrester (1000V class)	1.0	PRT-SPD-1000V — Higher-voltage SPD for large arrays.
	Installation & commissioning (large)	1.0	SRV-INSTALL-8000 — Labour for larger arrays (>25 kW).
	Transportation of logistics	1.0	SRV-TRANSPORT-STD — Delivery of equipment to site (Ghana proforma reference).

Site / manual items (confirm on site)

☐	Item	Notes
	Engineering / design drawings	String layout, single-line diagram, load study — not auto-generated.
	ECG / utility application fees	Grid-tied export or net-metering approvals vary by region.
	Scaffolding / height access	Multi-storey or steep roof — site-specific.
	Concrete ballast / ground mount civils	Ground-mount only; not in roof-mount BOM.
	Generator integration panel	Add when changeover feeds existing gen set beyond standard CO.

	Monitoring / Wi-Fi dongle	Inverter brand-specific; add from equipment catalog.
--	---------------------------	--

Commissioning sign-off

Installer / lead technician

Client representative

Date completed

Energy Precisions QC

Generated from sizing-driven BOM. Tick each line when delivered to site and installed. Final string design (VD, Isc) remains the supervising engineer's responsibility.